Exact values

1 Zero

2

a $x=\frac{\pi}{4}$

b $x=\frac{\pi}{3}$

c $x=\frac{\pi}{4}$

d $x=\frac{\pi}{6}$

3

a $x=\frac{\pi}{3},-\frac{2\pi}{3}$

b $x=\frac{\pi}{6},-\frac{5\pi}{6}$

c $x=\frac{\pi}{12},\frac{5\pi}{12},-\frac{11\pi}{12},-\frac{7\pi}{12}$

d $x=-\frac{\pi}{3},\frac{2\pi}{3}$

4

a $\theta=0,\pi,-\pi$

b $\theta=0,\pi,2\pi,3\pi,4\pi,-\pi,-2\pi,-3\pi,-4\pi$

c $\theta=\frac{\pi}{6},-\frac{\pi}{6}$

d $\theta=\frac{7\pi}{6},\frac{11\pi}{6},-\frac{5\pi}{6},-\frac{\pi}{6}$

e $\theta=0,2\pi,-2\pi$

f $\theta=\pm\frac{\pi}{6},\pm\frac{5\pi}{6},\pm\frac{7\pi}{6},\pm\frac{11\pi}{6},\pm\frac{13\pi}{6},\pm\frac{17\pi}{6},\pm\frac{19\pi}{6},\pm\frac{23\pi}{6}$

5

a $x=\frac{\pi}{6},\frac{5\pi}{6}$

b $x=\frac{5\pi}{6},\frac{7\pi}{6}$

c $x=\frac{2\pi}{3},\frac{4\pi}{3}$

d $x=\frac{\pi}{4},\frac{7\pi}{4}$

e $x=\frac{\pi}{4},\frac{5\pi}{4}$

f $x=\frac{7\pi}{6},\frac{11\pi}{6}$

g $x=\frac{\pi}{4},\frac{5\pi}{4}$

h $x=\frac{3\pi}{4},\frac{7\pi}{4}$

i $x=\frac{\pi}{12},\frac{\pi}{4},\frac{3\pi}{4},\frac{11\pi}{12},\frac{17\pi}{12},\frac{19\pi}{12}$

j $x=\frac{2\pi}{3},\frac{4\pi}{3}$

k $x=\frac{2\pi}{3},\frac{4\pi}{3}$

l $x=\frac{3\pi}{2}$

m $x=\frac{2\pi}{3}$

n $x=\frac{\pi}{3},\frac{5\pi}{3}$

o $x=0$

p $x=\pi$

q $x=\frac{\pi}{3},\frac{2\pi}{3},\frac{4\pi}{3},\frac{5\pi}{3}$

6

- **a** $x = \frac{\pi}{2}$
- **b** $x = \pi$
- **c** $x = \frac{\pi}{6}, \frac{5\pi}{6}$
- **d** $x = \frac{5\pi}{4}, \frac{7\pi}{4}$
- **e** $x = \frac{2\pi}{3}, \frac{4\pi}{3}$
- **f** $x = \frac{5\pi}{6}, \frac{11\pi}{6}$
- **g** $x = \frac{\pi}{3}, \frac{2\pi}{3}$
- **h** $x = \frac{\pi}{3}, \frac{5\pi}{3}$
- **i** $x = \frac{\pi}{6}, \frac{7\pi}{6}$
- **j** $x = \frac{\pi}{8}, \frac{3\pi}{8}, \frac{9\pi}{8}, \frac{11\pi}{8}$
- **k** $x = \frac{5\pi}{18}, \frac{11\pi}{18}, \frac{17\pi}{18}, \frac{23\pi}{18}, \frac{29\pi}{18}, \frac{35\pi}{18}$
- **l** $x = \frac{\pi}{12}, \frac{7\pi}{12}$
- **m** $x = \frac{\pi}{2}, \frac{7\pi}{6}$
- **n** $x = \frac{3\pi}{4}, \frac{7\pi}{4}$
- **o** $x = \frac{11\pi}{6}, \frac{\pi}{2}$
- **p** $x = \frac{4\pi}{3}$
- **q** $x = \frac{\pi}{2}, \pi$
- **r** $x = \frac{3\pi}{2}, \frac{\pi}{6}$

7

- **a** $\theta = 0, 2\pi$
- **b** $\theta = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{3\pi}{2}$
- **c** $\theta = 0, \frac{2\pi}{3}, \pi, \frac{5\pi}{3}, 2\pi$
- **d** $\theta = \frac{\pi}{4}, \frac{5\pi}{4}, \frac{3\pi}{4}, \frac{7\pi}{4}$

Other radian values

8 Two solutions

9

- **a**
  - **i** $\theta = 0.35$
  - **ii** $\theta = 0.35, 2.79$
- **b**
  - **i** $\theta = 0.44$
  - **ii** $\theta = 0.44, 5.84$
- **c**
  - **i** $\theta = 0.70$
  - **ii** $\theta = 0.70, 3.84$
- **d**
  - **i** $\theta = 1.33$
  - **ii** $\theta = 1.33, 4.95$
- **e**
  - **i** $\theta = 1.45$
  - **ii** $\theta = 1.45, 1.70$
- **f**
  - **i** $\theta = 0.66$
  - **ii** $\theta = 0.66, 3.80$
- **g**
  - **i** $\theta = 0.84$
  - **ii** $\theta = 0.84, 5.45$
- **h**
  - **i** $\theta = 0.09$
  - **ii** $\theta = 0.09, 3.05$
- **i**
  i$\theta =$ ii$\theta =$
  $0.23 \quad 0.23, 2.92, 3.37, 6.06$

10 $x = \pi - p$

11 $x = 2\pi - n$

12 $x = \pi + t$